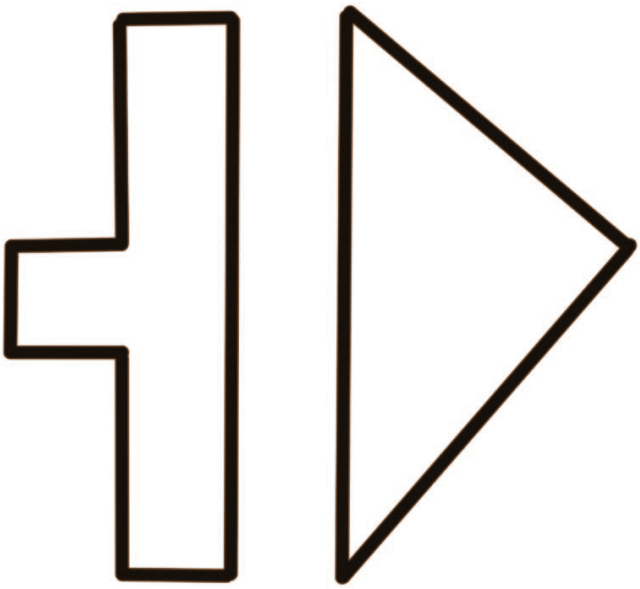
Brake-Inator

Inator-Makers



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05.01.2025

GEEN 1400-101

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Problem Formulation

Our question of interest is: how might we improve wheelchair brakes? Currently, the brakes available for manual wheelchairs are lacking, in multiple ways. They are unreliable, not very effective, and can cause strain for users with limited arm mobility and strength. Our product aims to reduce each of these problems. According to BODY FIT PHYSIOTHERAPY (*source 5)* : Many wheelchair users deal with wrist and arm pain caused by them having to brake the wheel chair manually by either placing their arms in the wrong position or spreading them to a long extension. According to nu motion *(source 6*): Braking a wheelchair becomes more difficult as time passes, since it damages the upper body area of the individual every time the person brakes the wheelchair. We're producing a product that is more reliable and effective while the person on the wheelchair is braking their wheelchair. Our idea is to attach a disk mountain bike brake into a wheelchair, since it is easy to manufacture and control without causing strain on the user's arms. With this product we are targeting wheelchair users with limited arm mobility to make it easier and safer for them to use wheelchairs. According to *Wheel Locks (source 3):* Unlocking Safety and Function, wheelchair brakes are named misleadingly. Wheelchair brakes are not used for braking, they simply “keep a stationary wheelchair stationary”. This is one issue we are aiming to solve. Our brakes will be adjustable, so that while in motion, a wheelchair user will be able to gradually slow down with a simple twist of a dial, rather than using arm strength to grip the wheels to slow themselves down.

Another issue currently faced by wheelchair users is that fully automated wheelchairs are extremely difficult to obtain. While our product won’t fully automate the wheelchair, it provides a system that can greatly improve the quality of someone with a manual wheelchair.

Design Requirements

Our design must be able to fully stop the motion of a wheelchair within 15 feet of initiation of the brake when moving at 10 miles per hour. This measurable standard comes from our consideration of the standards used for measuring the efficacy of bike brakes in regulation *16 CFR § 1512.18 - Tests and test procedures.16 CFR § 1512.18 - Tests and test procedures*, which states a brake must be able 2 to stop a bike fully within 15 feet while moving at 15 miles per hour *(source 7*). Our speed is slightly lowered from this standard due to the fact that the average speed of a manual wheelchair user is about 2 miles per hour (*source 8*).

Our design must be adaptable to many different wheelchairs, not only the wheelchair we acquired for testing. We want to make our design easily able to be attached to manually pushed wheelchairs of all different brands so the system should be able to universally fit onto any type of wheelchair spokes.

Our design must be safe for the user. Our goal is to create a safer wheelchair braking system that doesn't require as much effort from the users.

Our design must not interfere with any other functions of the wheelchair, such as the rotation of the wheel or the manual pushing of the wheelchair.

Our design must include an electrical component that does most if not all of the work of breaking the wheelchair, and improve upon currently available designs.

Initial Brainstorming

Bike Brake: The bike brake was one of our first ideas. This idea entailed mounting a typical bike brake that clamps on either side of the wheel as seen in figure 1. The user would then either clamp onto the handles, like on a bike, or adjust a slider to control the level of braking. We thought that this would be effective at braking, and would increase the users control over the braking. However, we also knew that this would likely get in the way of moving the wheelchair, and would be difficult to mount, and might not mount to all wheelchairs.

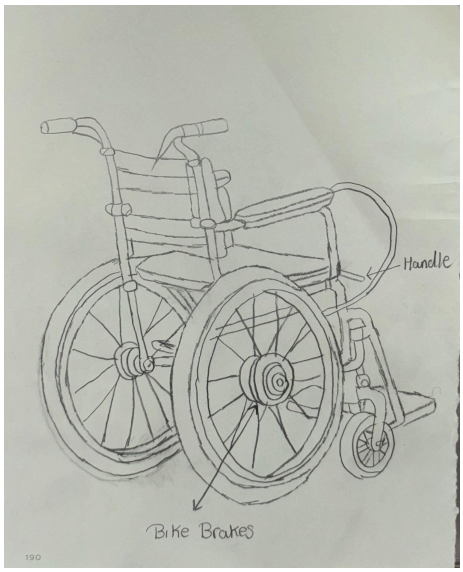


Figure 1 . Sketch of bike brakes mounted to wheelchair.

Disk Brake: The disk brake method is our final plan. We will attack disk brakes to the inside of the wheels, on both sides, using a machined piece to mount them as seen in figure 2. The disk braking system will have cords going from the brakes up to the arm of the wheelchair, where we will use a motor attached to a flywheel, possibly attached to a gear box, to pull the brake cords. The user will spin a dial which will engage this system, allowing for easy, adjustable braking.



Figure 2. Sketch of bike disk brakes placement on the wheelchair.

Spring System: This idea is a modified version of the current locking system typically used for brakes. The idea was to use a spring loaded brake system that would hold the brakes locked using springs, as seen in figure 3. To unlock the brakes the springs would be compressed and would release the stopper. This system would improve upon the current standard brakes by making it so that brakes could be easily engaged, and would not go out of alignment, however, the brakes would not be adjustable which was a key component of our ideal design. Not being adjustable, along with complex and high level mechanics involved in designing such a system meant that this design was not ideal for this project.

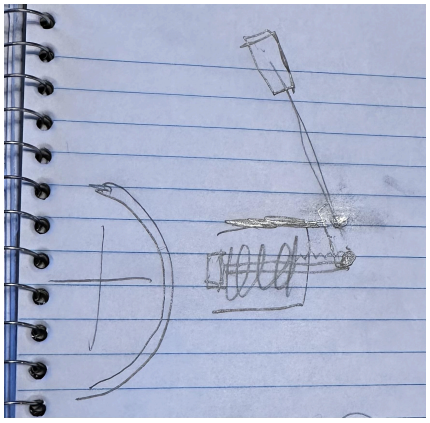


Figure 3. Simple sketch of the spring system.

Vice Grip: This idea would be a simple solution to most of the issues of current wheelchair brakes. The design is inspired by a vice grip that locks into place, and is adjustable, as seen in figure 4. The idea was to replicate this functionality into wheelchair brakes, which would mean that the tightness of the brakes would be adjustable, and disengaging the brakes would be much easier through the use of a release lever (similar to vice grips). This idea was one of our first ideas, and did not solve the issue of needing adjustable braking, and would be difficult to implement with arduino components, so we decided to leave this idea early on and possibly use its key mechanism as inspiration for other designs.

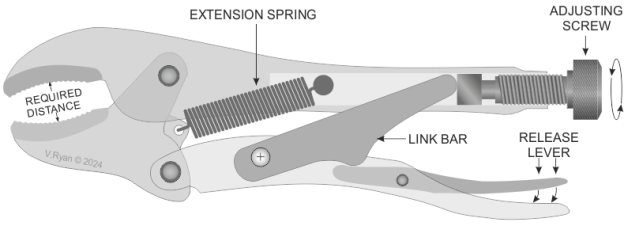


Figure 4. Vice grip internal locking mechanism.

Lift Off the Ground: This idea was one we almost immediately decided was unfeasible. It would include metal legs near the wheels of the wheelchair that, when engaged, would push the wheelchair off the ground, to stop the wheels from spinning, and essentially turning it into a regular chair as seen in figure 5. While this would have

been effective at stopping the wheelchair, it wouldn't have worked in many situations, such as a hill.

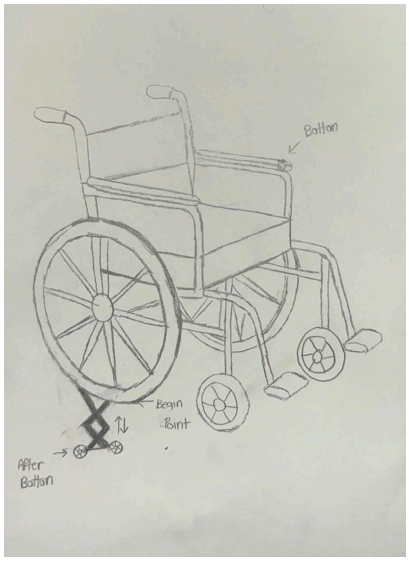


Figure 5 . Sketch of our ground lifting idea.

Idea Section

Reqirements	Ease of Use	Safety	Cost	Appearance	Accessibility	Braking Efficacy	Row Total	Normalized Value
Ease of Use		0	1	1	0	0	2	0.13
Safety	1		1	1	1	1	5	0.33
Cost	0	0		1	0	0	1	0.07
Appearance	0	0	0		0	0	0	0.00
Accessibility	1	0	1	1		0	3	0.20
Braking Efficac	1	0	1	1	1		4	0.27

Figure 6 . A pugh chart with our top six requirements for the project.

Reqirements	Weights	Disk Bike Br	Regular Bike Brakes	Adjustable Brake System
Ease of use	0.13	0	0	0
Safety	0.33	0	0	0
Cost	0.07	0	0	-1
Appearance	0.00	0	0	0
Accessibility	0.20	0	-1	0
Braking Efficacy	0.27	0	-1	-1
	Weighted total:	0	-0.4666666667	-0.3333333333

Figure 7 . A chart comparing our top three ideas with our top six requirements.

Our top idea was selected after we had started working on a different idea. The original idea was the spring break, and after working on it for a week, we decided that it wasn't going to work the way we had wanted it to. Some people in our group were also unsure of how it was actually going to work, so we switched to an idea we were all confident in. This idea was the disk brake solution. This solution is much more feasible, and will be overall more effective at accomplishing all of the issues we want to solve with our design.

The disk brake will be easily mounted onto the wheelchair with the use of machined parts and other materials, and will provide safe, effective braking. It will be

hooked up to a dial that will sit on the arm of the wheelchair. The user will spin the dial to increase or decrease the level of braking. There will also be a button that will induce automatic full force on the brake, to stop the wheelchair immediately. This will be important to make sure the system is as safe as possible, which was our top ranked requirement, as seen in figure 6. The disk brakes will be easy to mount to any wheelchair, and will provide the most effective gradual braking possible, as shown in figure 7.

Design Alterations

For our final design, we decided to use a disk brake system to effectively slow down the wheelchair. The design needed to have variable braking that effectively applied braking force, and be attached to the wheelchair without interfering with normal function.

Sub Function 1: Applying Braking Force

The first sub function is one of the more integral to the project. We chose to do this through the use of a disk brake system. This entails a disk and a caliper, both of which must be attached to the wheelchair. The disk rotates through the caliper, while the caliper is held in place. The brake cable, attached to the caliper, gets pulled by the electronics component, which then clamps the disk. The disk is attached to the wheel and rotates with it, so as the caliper clamps down on the disk and slows it down, it also slows down the wheel, effectively braking the entire wheelchair. One design alternative we considered was using a traditional bike brake system, one where there are clamps that directly clamp down on the wheels. We decided against this one, as we thought it could interfere with the normal wheelchair use, and wouldn't be as effective as the disk brake system. A second design alternative was a spring loaded braking system. This idea entailed two springs placed between two pieces of 3D printed rectangles, which would be anchored next to the wheel and would be passively locked. When engaged, a ratcheting system would allow the springs to extend to their natural length, pressing the 3D printed material into the wheel to apply force and friction to slow it down. We decided against this idea, as it would not have been easy to implement, and wouldn't be particularly effective.

Sub Function 2: Attaching to the Wheelchair

The second sub function is the system's way of attaching to the wheelchair. In our beta prototype our electronics box is attached to the wheelchair with fabric straps wrapped around the handles and has velcro attached to the back of the wheelchair. The calipers are attached to 3D printed caliper mounts which are two pieces each: the first piece is where the caliper is bolted on to, and the second piece completes the circular shape which wraps around the bar the mount attaches to, and holes on each piece are fed through with wire which is fastened around it to keep the mount in place.
Each of these pieces had design alternatives. For the electronics box, we first considered having it hang under the wheelchair, but we pivoted away from that after deciding the space under would limit the size we could make the box. We also considered using wire to hang the box off of the handles, but we thought the thicker fabric loops would provide better stability. For the caliper mounts, most of our prototyping involved the mounts being a single piece which snaps into place on the metal bars of the wheelchair. We eventually decided to separate it into two pieces, as the single piece gave us very little room for error in the prints, and tended to break after being attached and removed several times.

Sub Function 3: Variable Braking

Our third sub function is part of what allows our product to differentiate itself from traditional brakes. We needed the brakes to gradually slow down the user, rather than just lock the wheels in place. This was done through the use of servos and flywheels, as well as a potentiometer to control the level of braking. In the electronics box, we have two servos and two flywheels, attached to the servos. The brake cable is also attached to the flywheels, and connects to the calipers. The potentiometer is located on the arm of the wheelchair, and as the user spins it, it engages the servos and flywheels. The flywheels turn a certain amount depending on the angle of the potentiometer. If the

flywheel is turned less, the brake cable is pulled less, meaning there is less braking force on the disk. We considered using a ratcheting system for variable braking, through the use of vice grips. This would have worked with the spring loaded brake system. We decided against this idea as we thought that it wouldn't provide as much variability as we wanted, and would be difficult to make it electronically engaged. We also discussed using the handles typically seen on bikes. We tested this and it worked well for braking. We decided not to use this, however, because we wanted it to be electronically engaged, and we wanted it to be as accessible as possible for people with limited upper body mobility.

Alpha Prototype

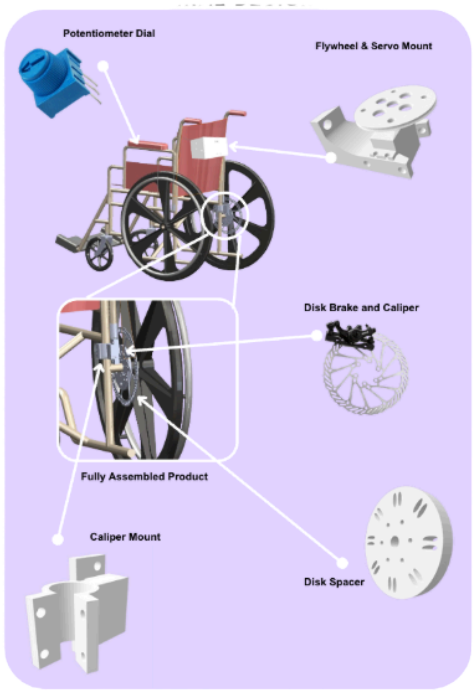


Figure 8. Shows the final design of our product highlighting where everything goes on the wheelchair.

Our alpha prototype consists of five main components, as shown in Figure 8. The first component is the disk brake and caliper assembly. The disk is mounted to both wheels and a

spacer, which ensures rotation is in unison with the wheel. The caliper, which contains brake pads on either side of the disk, is fixed to a mount using two bolts, allowing it to remain stationary while the disk rotates through it.

The disk spacer creates the necessary clearance between the wheel and the wheelchair frame, enabling the proper caliper alignment without interfering with the wheelchair's normal function. The caliper mounts are attached to the chair's metal frame near the wheel, securing the calipers in a fixed position to ensure consistent braking performance.

The electronics box, located on the back of the wheelchair, houses the servos, flywheels, Arduino, and breadboard. The servo and flywheel system acts as a digital analog to a manual bike brake lever. Each servo, held in place by a dedicated mount, turns a flywheel that pulls the brake cable. The servos are activated based on the input angle from a potentiometer dial located on the sidearm of the chair, which measures the user's intended breaking force via how much they turn the dial.

Engineering Analysis

One of the key functions of our product involves brake cables being pulled by servo motors. To select an appropriate servo, we first needed to determine the torque required to pull the cables and fully engage the brake calipers. Using the torque equation $\tau = r \times F$, where r is the radius and F is the applied force. We found the required force to close the calipers of 20 N and a radius of 6 cm, from these numbers we calculated the necessary torque to be 1.2 Nm. Based on the necessary torque, we selected a servo rated at 35 kg*cm, which provides approximately 3 Nm of torque. This is well above our minimum threshold and ensures reliable braking.

Beta Prototype



Figure 9: A spring contraption made from PLA and two springs.

Our beta prototype was the first design idea we had, which changed later. This design entailed a spring loaded braking system. The piece seen in figure #9 shows two sturdy springs attached to two rectangular pieces, 3D printed in PLA filament. This system would be placed next to the wheels on the wheelchair. They would be passively locked using some sort of ratcheting system. When engaged, the springs would be extended to their natural length, and would push against the wheels with the force of the springs, slowing the wheels down.

Manufacturing Plan

Raw Materials, Parts or Components	Unit Cost	Quantity	Total Cost
3D Printer	$0	1	$0
Acrylic Sheet	$5	1	$5
Arduino Kit	$0	1	$0
Bolts, Nuts, Washers Pack	$6	1	$6
Disk Brake Kit	$31	1	$31
Filament	$0		$0
Galvanized Wire	$7	1	$7
Laptop			
Long Wheel Bolts	$6	2	$12
Metal Strapping	$9	1	$9
Poster	$30	1	$30
Poteniometer Dial	$0	1	$0
Protoboard	$0	1	$0
Servo	$29	2	$58
Super Glue	$0	1	$0
Wheelchair	$0	1	$0
Wires	$0	15	$0
			$158

Figure 10. Image of our Bill of Materials (BOM).

The first step of the assembly of our Alpha prototype is attaching the flywheel and our disk from the disk brakes. The wheels were removed from the wheelchair, fitted with new bolts, then had the disk and spacer fitted with the bolt through them, then metal strapping was wrapped and tightened around three parts so that they are securely fastened together. Next after the assembly of the wheel is the attachment of the caliper mounts. First, each mount has the calipers attached to them with two short bolts that don't go past the width of the mount, caliper, and nut together. When looking at the wheelchair from behind, the right caliper mount is lined up with the gap within the caliper over the disk brake, and is attached to the very top of the long bar going straight up and down next to the wheel. The left mount is lined up and attached in the same way but this time on the bottom of the long bar.The next step is the assembly of the electronics box. Each servo must be snapped into one of the two servo mounts, and have a flywheel attached to it. Each mount is then lined up with its respective holes on the 3D printed electronics box and has a long bolt fitted through the holes and locked in with a nut. Next, hinges are screwed into the side of the box and the laser cut acrylic top so that the two are

connected.The arduino is connected with its power cable to a laptop and our code (Fig 11) is uploaded.The Arduino and proto board attached together are then connected to the acrylic top with double sided gorilla tape, along with the two 9 volt batteries powering the servos. Then wires are hooked up to the servos and protoboard in the connections shown (Fig 12). Next, double sided gorilla tape is then used to attach the arduino battery pack to the back of the box, and its cable is fed through the hole on the back of the box and connected from the arduino to its battery pack. The final electronic wiring for the electronics box is complete when the potentiometer wiring is fed through the hole on the back of the electronics box. Each brake cable is fitted into its black cable housing and then fed through the holes on the shooter sides of the box and through the servo mount hole, and the inner metal brake cable is routed around the flywheels and tied through the holes of the flywheels with zip ties. Velcro is then cut and the hooked side is attached to the back of the wheelchair while the toothed side is attached to the back of the electronics box. Last for the electronics box, the fabric straps are fed through the holes on each side and the box is hung from the handles on the back of the wheelchair, and the velcro parts are connected together. The brake cables are then each fed through to the caliper and tightened into place with a hex wrench. Finally, the potentiometer is attached to the horizontal bar in front of the right arm rest, and wire is connected through the holes on the potentiometer mount to tighten it into place.

```
#include <Servo.h>
Servo myservo;
Servo oppServo;
int pos = 0;
const int readFrequency = 15;

void setup() {
 myservo.attach(9,500,2500);
 oppServo.attach(10,500,2500);
 pinMode(A0,INPUT);
 Serial.begin(9600);
}

void loop() {
  int potVal = analogRead (A0);
  Serial.println(potVal);

  pos=map(potVal,10,1015,0,180);
  myservo.write(pos);
  oppServo.write(pos);

  delay(50);
}
```

Figure 11 This is an image of our Arduino code.

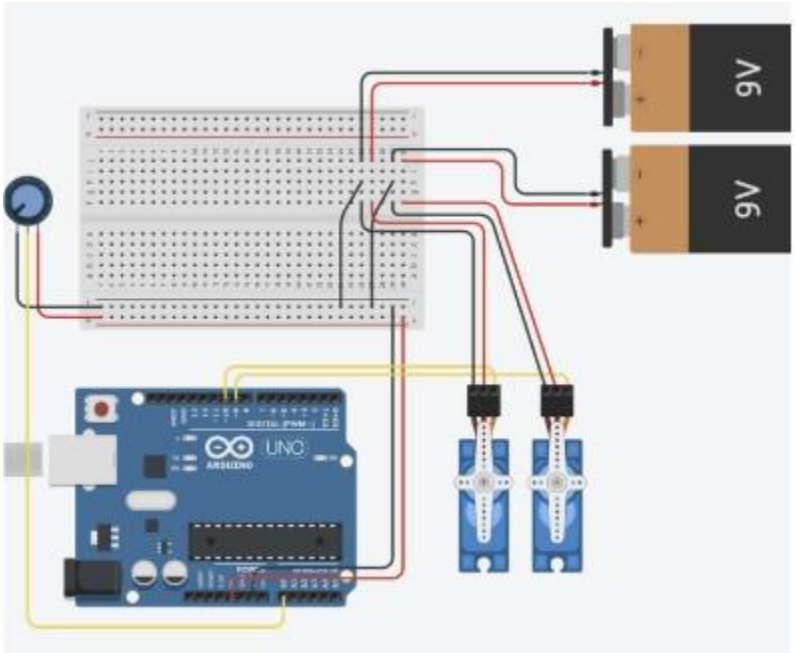


Figure 12 Image of our Arduino and Protoboard wiring.

Testing and Validation

A key performance requirement for our braking system is that the wheelchair must be able to come to a complete stop within 15 feet when traveling at a speed of 10 miles per hour. This specification is intended to ensure safe and reliable;e stooping in real-world scenarios, where quick response and braking efficiency are critical. Since there are no testing standards on wheelchair breaks we developed this number by using the standard for bike brakes.

To evaluate this requirement, we have developed a testing plan in which the wheelchair will be accelerated at a speed of 10 mph and the braking system will be engaged. The distance from the point of brake activation to a complete stop will be measured. Multiple trials will be performed to ensure consistency and account for any variability due to surface conditions or brake calibration. Success will be determined by whether the average stopping distance across all trials is less than or equal to 15 feet. If the prototype meets this standard, it will validate the effectiveness of our disk braking system. These tests will also help identify any adjustments needed in brake calibration, servo power, or component alignment to improve the performance.

A second requirement is that our servos must generate 1.2 Nm of torque to fully engage the brake calipers. This value was calculated using the equation $\tau = r \times F$, where the force required to close the caliper is 20 N and the flywheel has a 6 cm radius. To validate this, we will construct a test setup where the servo pulls a brake cable via a flywheel. A spring scale will measure the force exerted on the cable, and the radius of the flywheel will be recorded to calculate the torque. We will run this test for multiple trials to confirm consistency. If the measured torque meets or exceeds 1.2Nm in all trials, the amount of torque would be successfully validated.

Inclusive Analysis

One design aspect for improving the inclusivity of our product is simplifying the assembly. Our final product, as presented at expo, was not easy to install. To address this, future iterations will focus on making assembly more user-friendly. One proposed idea for a solution is to manufacture a custom wheel with a pre-installed disk brake, allowing users to simply replace the existing wheel with minimal effort. Another option is to offer an assembly service in which the braking system is installed directly onto the wheelchair by a trained professional.

Additionally, we planned to improve the accessibility of the control interface by addressing the potentiometer to better accommodate users with limited dexterity. This may involve incorporating larger more ergonomic dial or alternative input methods. We also intend to allow flexible mounting locations for the dial, enabling users to position it wherever is most comfortable or practical for their individual needs.

As we continue to refine the product, obtaining direct feedback from wheelchair users will be essential. Designing with input from wheelchair users ensures that the final product meets real-world needs and avoids the able-bodied assumptions driving accessible designs.

Risk Analysis

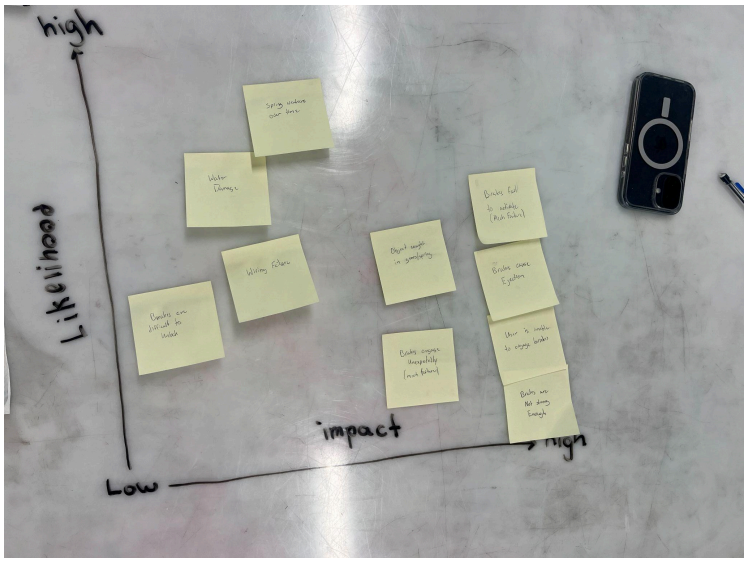


Figure 13. This is an image of our risk analysis graph.

Our highest impact and most probable risk, illustrated in Figure 13, was brake failure. To mitigate this risk, we implemented mountain bike disk brakes, which are specifically designed for high impact stopping scenarios. These brakes offer the reliability and stopping power necessary to ensure effective braking for the wheelchair.

Another major risk we identified was the potential for foreign objects to become entangled in the spring or braking mechanism. Following our Preliminary Design Review (PDR), we underwent a significant design shift from a spring system to a disk brake configuration. As a result this risk became less prominent. However, for future manufacturing, we plan to further reduce this risk by designing the system to be more preassembled. This approach would minimize the number of exposed components, thereby reducing the chance of interference with the braking system.

Stakeholder Feedback and Future Work

As part of our research on the prototype, we posed five specific questions at the engineering expo:

How easy is it to use the brake system? Would you say it is user-friendly?

Is the user able to reach the dial easily, or is it ergonomic during operation?

Do you have any complaints regarding the placement of the electronic box, and its potential failure?

Would it be okay for you to use this braking system on a daily basis and feel safe?

What measures do you think would make the product less complicated, or simpler to equip?

As for the provided feedback, most respondents replied positively, noting useful considerations. Numerous stakeholders expressed appreciation for the design's functionality, although it was not functioning properly at the time, as an assistive device for manual wheelchair users who do not have access to motorized wheelchairs. Some participants regarded the placement of the dial on the armrest as convenient and easy to reach, while others argued that it needed to be movable to accommodate placement for varying levels of physical ability or wheelchair shape. This feedback demonstrated an inclination towards customizable mounting solutions that allow optimal positioning of the dial regarding personal comfort and accessibility.

In general, the comments captured most, if not all aspects of the requirements, as they suggest that the system could prove to increase safety and ease of use for a broad range of users of manual wheelchairs and focus on more design personalization flexibility.

While enhancing the wheelchair braking system, the safety features have been noticed that need improvement. Safety, as well as accessibility, durability is also weakened . When focusing on improving the design, replacing 3D printed parts with aluminum or stainless steel

comes to mind as a priority. Ensuring that the 3D printed parts are replaced with metal, the construction would allow the mounting brackets and grab calipers mechanical components to outlive daily stress from the environment. These materials will guarantee enabling user handling with degradation in performance or safety.

To boost reliability for users depending on the braking system throughout the day, switching to travel focused modes will enable changing the scenario. Stakeholders have also pointed out the focus on battery life, making them rechargeable will boost reliability especially for people who need to utilize the braking system frequently during the day, such as when outdoors.

Furthermore, we hope to increase the ergonomic accessibility of the control interface by upgrading the potentiometer to be more accommodating for users with restricted mobility. This may require a large and more comfortable dial to be added, or we may consider other forms of input such as push buttons. We also plan to allow users to place the dial wherever is most convenient for them, offering greater autonomy in mounting configuration for greater comfort in relation to their wheelchair.

In terms of accessibility, we also aim to offer optional installation services. Some users may be unable to attach the system themselves, so buying a fully assembled system could be a better option for these users. To accommodate them, trained technicians would be available to assist buyers with proper installation to ensure the system is operational and secured correctly. We also hope to partner with insurance providers to make the product affordable and accessible to a larger segment of the population. Our ultimate aim is for the braking system to be included under mobility aid plans, as those who cannot afford a fully electric wheelchair would realistically need them.

In the end, our braking system functions as an economical and efficient wheelchair brake from hand-operated till motorized. With these design changes, we passionately strive to improve the dependability, safety, and availability of the system enhancing independence and ease to more users at minimal cost.

Team Summary

Julia came up with the concept for the bike brake disk and assisted in the initial stages of the assembly. She bought the disk brakes and was involved in their integration into the wheelchair. Additionally, she participated in preparing the expo poster as well as the final project technical report. Julia did the bulk of the 3D models for the mounting structure for the disk and caliper.

Grace participated in forming the concept of disk brakes. She assisted with the construction of the disk brakes. She was also active in designing the expo poster and writing the technical report that was prepared for the project. Additionally, Grace did some laser cutting for the electronic box and the logo was placed on it, which she then assembled to the 3D printed electronic box.

Rameez contributed to the design of the disk brake idea and helped in the construction of the disk brakes. She also helped in preparing the expo poster and in writing the final report in the project's Technical Report. Rameez also assisted Grace with the measurement of the electronic box and with the assembly of the laser cut part to the 3D printed electronic box.

After meeting with an engineer, Timo strategized on how to make the disk brake system adjustable and linkable to an Arduino. After further evaluating his work, he conducted research on the effectiveness of mountain bike disk brakes and used that data to evaluate the efficacy of

our project. Even though most of the electronic components were Timo's responsibility, he also contributed to the building aspect of the project. He was also involved in the writing of the final technical report.

Stephen worked on the Arduino and did research with Timo. He mostly contributed towards the electronic parts of the project with Timo, and had a slight hand in helping out with the construction of the wheelchair. Other than that, Stephan also helped with the expo poster.

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Appendices

How might we…

- repurpose garden water/rain/snow
- make safety resources more accessible
- Salt sidewalks better
- Make pulse ox less racist
- Sanitize rooms with a UV light
- Minimize drinks being roofied
- Detect wildfires better
- Detect sugar in drinks for diabetes

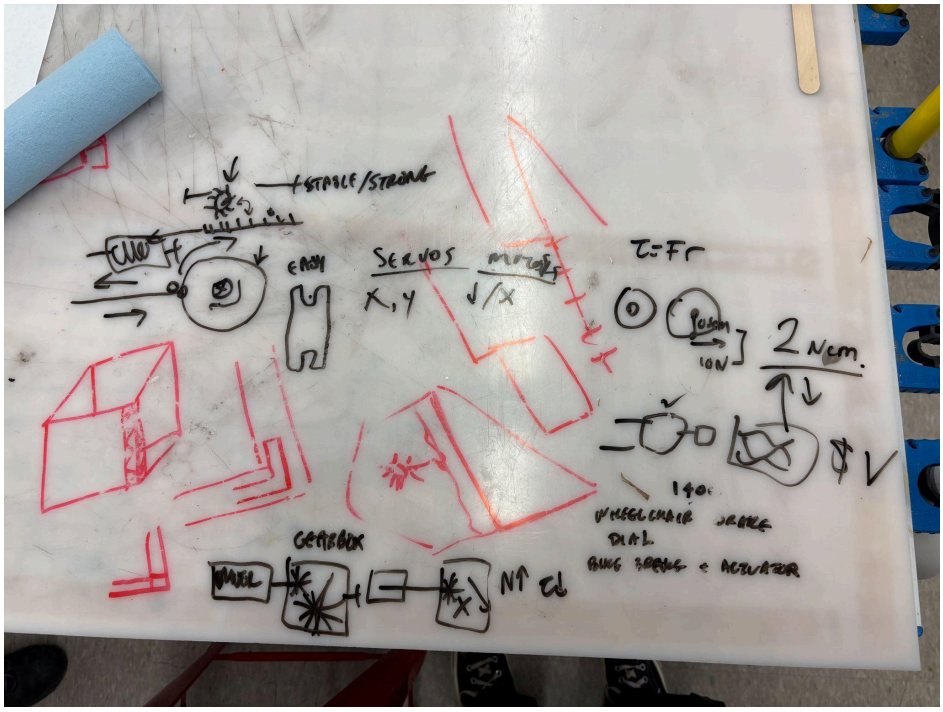


Figure 14. Sketch of how we attach the brake cable to a flywheel which is then pulled by a motor and attached to an Arduino interface.